



ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Section-II (Short Questions)

Marks=20

- Linear operator A has n distinct eigenvalues and n corresponding eigenfunctions: $A\psi_i = \lambda_i \psi_i$. Show that the n eigenfunctions are linearly independent. Do not assume A to be Hermitian.
- A triangular wave is represented by

$$f(x) = \begin{cases} x, & 0 < x < \pi \\ -x, & -\pi < x < 0. \end{cases}$$

Represent $f(x)$ by a Fourier series.

- Show that

$$\mathcal{F}[f^*(-t)] = F^*(\omega),$$

where $F(\omega) = \mathcal{F}[f(t)]$.

- Using

$$\Gamma(z) = \lim_{n \rightarrow \infty} \frac{1.2.3 \dots n}{z(z+1)(z+2) \dots (z+n)} n^z, \quad n \neq 0, -1, -2, \dots;$$

show that

$$\Gamma(z+1) = z\Gamma(z).$$

- Evaluate $\int_{-\infty}^{+\infty} x^3 e^{-x^2} H_n(x) H_n(x) dx$.

Section-III

Marks=30

- Show that

$$G(x, t) = \begin{cases} x, & 0 \leq x \leq t, \\ t, & t \leq x \leq 1, \end{cases}$$

is the Green's function for the operator $L = -\frac{d^2 y}{dx^2}$ and the boundary conditions $y(0) = 0, y'(1) = 0$.

- Verify the Dirac delta function expansions

$$\delta(1-x) = \sum_{n=0}^{\infty} \frac{2n+1}{2} P_n(x),$$

$$\delta(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{2n+1}{2} P_n(x).$$

- Expand x^r in a series of associated Laguerre's polynomials $L_n^{(k)}$, with k fixed and n ranging from 0 to r (or to ∞ if r is not an integer).

- Show, by direct differentiation, that,

$$J_\nu(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s! \Gamma(s+\nu+1)} \left(\frac{x}{2}\right)^{\nu+2s},$$

satisfies the two recurrence relations

$$J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x),$$

$$J_{\nu-1}(x) - J_{\nu+1}(x) = 2 \frac{dJ_\nu(x)}{dx}.$$

- Evaluate

$$\int_{-1}^1 (1+x)^a (1-x)^b dx,$$

in terms of the beta function.



UNIVERSITY OF THE PUNJAB

Sixth Semester - 2018

Examination: B.S. 4 Years Programme

M.M./P

Roll No. 014865

PAPER: Mathematical Methods of Physics-II
Course Code: PHY-307 Part - II

TIME ALLOWED: 2 Hrs. & 45 Mints.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Section-II (Short Questions)

Marks=20 ¹²
(4x5)

1. The functions $u_1(x)$ and $u_2(x)$ are eigenfunctions of the same Hermitian operator but for distinct eigenvalues λ_1 and λ_2 . Prove that $u_1(x)$ and $u_2(x)$ are linearly independent.
2. A different sawtooth wave is described by

$$f(x) = \begin{cases} -\frac{1}{2}(\pi + x), & -\pi \leq x < 0 \\ \frac{1}{2}(\pi - x), & 0 < x \leq \pi. \end{cases}$$

Show that $f(x) = \sum_{n=1}^{\infty} \frac{\sin(nx)}{n}$

✗ Show that

$$\mathcal{F} \left[\frac{d^n}{dt^n} f(t) \right] = \left[\frac{d^n}{dt^n} f(t) \right]^T(\omega) = (-i\omega)^n \mathcal{F}[f(t)] = (-i\omega)^n [f(t)]^T(\omega).$$

✗ Use mathematical induction to show that

$$J_n(x) = (-1)^n x^n \left(\frac{1}{x} \frac{d}{dx} \right)^n J_0(x).$$

✗ Show that

$$\int_{-\infty}^{+\infty} x^2 e^{-x^2} H_n(x) H_n(x) dx = \pi^{1/2} 2^n n! \left(n + \frac{1}{2} \right).$$

Section-III

Marks=30 ²⁴
(6x5)

1. Find the Green's function for

$$\frac{d^2 y}{dx^2} + k \frac{dy}{dx} = f(x),$$

subject to the initial conditions $y(0) = y'(0) = 0$, and solve this ODE for $x > 0$ given $f(x) = \exp(-x)$.

2. A function $f(x)$ is expanded in a Legendre series $f(x) = \sum_{n=0}^{\infty} a_n P_n(x)$. Show that

$$\int_{-1}^{+1} |f(x)|^2 dx = \sum_{n=0}^{\infty} \frac{2a_n^2}{2n+1}.$$

✗ Show that

$$\int_0^{\infty} e^{-x} L_l(x) L_m(x) dx = \delta_{lm},$$

where $L_l(x)$ and $L_m(x)$ are Laguerre's polynomials.

✗ Derive the recurrence relations

$$\Gamma(z+1) = z\Gamma(z),$$

from the Euler integral

$$\Gamma(z) = \int_0^{\infty} e^{-t} t^{z-1} dt.$$

where z is a positive real number.

$$\Gamma'(z) = -\frac{1}{z} - \frac{1}{2z^2} - \frac{1}{6z^3} - \dots$$

✗ Show that

$$\Gamma(k+1) \Gamma\left(\frac{1}{2} + k\right) = \frac{\sqrt{\pi}}{2^{2k}} \Gamma(2k+1),$$

where k is an integer.



UNIVERSITY OF THE PUNJAB

Sixth Semester - 2017

Examination: B.S. 4 Years Programme

Roll No. 012879

PAPER: Mathematical Methods of Physics-I
Course Code: PHY-307

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Section-II (Short Questions)

Marks=20

1. Show that

$$\int_0^{\infty} e^{-x^4} dx = \Gamma\left(\frac{5}{4}\right)$$

✓ 2. Find the Fourier series of $f(x) = x^3$ for $0 < x \leq 2$.

3. Show that the linear operator $L = \frac{d^2}{dx^2}$ is self-adjoint, and determine the required boundary conditions for the operator to be Hermitian over the interval t_0 to $t_0 + T$.

✓ 4. Using a Rodrigues formula $(P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n)$, show that the $P_n(x)$ are orthogonal and that

$$\int_{-1}^{+1} P_n(x) P_m(x) dx = \frac{2}{2n+1}$$

5. Show that

$$\mathcal{L}\left\{\frac{\sin kt}{kt}\right\} = \frac{1}{k} \cot^{-1}\left(\frac{s}{k}\right)$$

Section-III

Marks=30

1. Evaluate

$$\int_0^{\infty} e^{-ax} J_0(bx) dx, \quad a, b > 0.$$

✓ 2. ✓ Show that

$$G(x, t) = \begin{cases} x, & 0 \leq x < t, \\ t, & t \leq x < 1, \end{cases}$$

is the Green's function for the operator $L = -\frac{d^2}{dx^2}$ and boundary conditions $y(0) = 0, y'(1) = 0$.

✓ 3. ✓ Prove that

$$\left(2x - \frac{d}{dx}\right)^n 1 = H_n(x),$$

also verify that

$$a_n \psi_n(x) = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx}\right) \psi_n(x) = n^{1/2} \psi_{n-1}(x),$$

$$a_n^\dagger \psi_n(x) = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx}\right) \psi_n(x) = (n+1)^{1/2} \psi_{n+1}(x),$$

where

$$\psi_n(x) = e^{-x^2/2} \frac{H_n(x)}{(2^n n! \pi^{1/2})^{1/2}}.$$

✓ 4. Derive the recurrence relation

$$H_{n+1}(x) = 2x H_n(x) - 2n H_{n-1}(x)$$

5. Show that the Fourier one transform of e^{-at} is

$$g(\omega) = \sqrt{\frac{2}{\pi}} \frac{1}{\omega^2 + a^2}$$



UNIVERSITY OF THE PUNJAB

Sixth Semester 2015
Examination: B.S. 4 Years Programme

Roll No. 7698

PAPER: Mathematical Methods of Physics-II
Course Code: PHY-307

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Section-II (Short Questions)

Marks=20

1. If Ψ is a solution of Laplace's equation $\nabla^2 \Psi = 0$, show that $\Psi_1 = \frac{\partial \Psi}{\partial x}$ is also a solution.

2. A triangular wave is represented by

$$f(x) = \begin{cases} x, & 0 < x < \pi \\ -x, & -\pi < x < 0 \end{cases}$$

Represent $f(x)$ by a Fourier series.

3. The functions $u_1(x)$ and $u_2(x)$ are eigenfunctions of the same Hermitian operator but for distinct eigenvalues λ_1 and λ_2 . Prove that $u_1(x)$ and $u_2(x)$ are linearly independent.

4. Prove that

$$\left(2x - \frac{d}{dx}\right)^n 1 = H_n(x)$$

(Hint: Use mathematical induction)

5. A function $f(x)$ is expanded in a Legendre series $f(x) = \sum_{n=0}^{\infty} a_n P_n(x)$. Show that

$$\int_{-1}^{+1} |f(x)|^2 dx = \sum_{n=0}^{\infty} \frac{2a_n^2}{2n+1}$$

Section-III

Marks=30

1. Show that

$$\frac{\sin x}{x} = \int_0^{\pi/2} J_0(x \cos \theta) \cos \theta d\theta.$$

Hint: The definite integral $\int_0^{\pi/2} \cos^{2s+1} \theta d\theta = \frac{2 \cdot 4 \cdot 6 \cdots (2s)}{1 \cdot 3 \cdot 5 \cdots (2s+1)}$, may be useful.

2. Find the Green's function for the operator

$$Ly(x) = \frac{d^2 y}{dx^2} + y, \quad y(0) = 0, \quad y'(1) = 0.$$

3. Show that

$$\int_{-\infty}^{+\infty} e^{-x} L_m(x) L_n(x) dx = \delta_{mn},$$

where $L_m(x)$ and $L_n(x)$ are Laguerre polynomials.

4. Solve the initial-value problem using Fourier transform

$$\frac{\partial^2 u(x,t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 u(x,t)}{\partial t^2} \quad u(x,0) = f(x), \quad \frac{\partial u(x,t)}{\partial t} \Big|_{t=0} = 0,$$

where $f(x)$ is assumed localized (i.e. $\lim_{x \rightarrow \pm\infty} f(x) = 0$).

5. Show that

$$\Gamma(n+1) \Gamma\left(\frac{1}{2} + n\right) = \frac{\sqrt{\pi}}{2^{2n}} \Gamma(2n+1),$$

where n is an integer.

UNIVERSITY OF THE PUNJAB

Sixth Semester 2014

Examination: B.S. 4 Years Programme

Roll No.

PAPER: Mathematical Methods of Physics-I
Course Code: PHY-307

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Section-II (Short Questions)

1. Use mathematical induction to prove

$$J_n(x) = (-1)^n x^n \left(\frac{1}{x} \frac{d}{dx} \right)^n J_0(x).$$

Marks=20

2. A sawtooth wave is given by

$$f(x) = x, \quad -\pi < x < \pi.$$

Show that

$$f(x) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx.$$

3. The functions $u_1(x)$ and $u_2(x)$ are eigenfunctions of the same Hermitian operator but for distinct eigenvalues λ_1 and λ_2 . Prove that $u_1(x)$ and $u_2(x)$ are linearly independent.

4. Derive the recurrence relation

$$(1-x^2)P_n'(x) = nP_{n-1}(x) - nxP_n'(x).$$

5. Derive the recurrence relations

$$\Gamma(z+1) = z\Gamma(z),$$

from the Euler integral

$$\Gamma(z) = \int_0^{\infty} e^{-t} t^{z-1} dt.$$

Section-III

1. Show that

$$\int_{-1}^{+1} x^n P_n(x) dx = \frac{2^{n+1} (n!)^2}{(2n+1)!} = \frac{2^n n!}{(2n+1)!}.$$

2. Find the Green's function for the operator

$$Ly(x) = \left(\frac{d^2}{dx^2} + y(x) \right), \quad y(0) = 0, \quad y'(1) = 0.$$

3. Show that

$$\int_{-\infty}^{+\infty} x^2 e^{-x^2} H_n(x) H_n(x) dx = \pi^{1/2} 2^n n! \left(n + \frac{1}{2} \right).$$

(Useful recurrence relation $H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$)

4. From the convolution theorem show that

$$\frac{1}{s} f(s) = \mathcal{L} \left\{ \int_0^t F(x) dx \right\}, \quad \text{where } f(s) = \mathcal{L}\{F(t)\}.$$

5. Show that

$$\Gamma\left(\frac{1}{2} - n\right) \Gamma\left(\frac{1}{2} + n\right) = (-i)^n \pi,$$

where n is an integer

See my Notes "Zaib Butt's"
My YouTube Channel: "Physics ki duniya by Zaib Butt" Also Subscribe it & share
(For video lectures), always remembers in your prayers my parents and me.



UNIVERSITY OF THE PUNJAB

Sixth Semester 2013
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Mathematical Methods of Physics-
Course Code: PHY-307

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Section-II (Short Questions)

Marks=20

1. Prove that

$$\left(2x - \frac{d}{dx}\right)^n 1 = H_n(x).$$

P#200

2. Expand

$$f(x) = x^2,$$

Example P#136 mentes

for $-\pi \leq x \leq \pi$ in a Fourier series.

3. Prove that the product of two Hermitian operators is Hermitian; (The operator L is called Hermitian if $\int \phi_1^* L \phi_2 d\tau = \int (L \phi_1)^* \phi_2 d\tau$) if and only if the two operators commute.

4. Verify the Dirac delta function expansion

$$\delta(1-x) = \sum_{n=0}^{\infty} \frac{2n+1}{2} P_n(x).$$

5. Show that between any two consecutive zeros of $J_n(x)$ there is one and only one zero of $J_{n+1}(x)$ (Useful relations $\frac{d}{dx}(x^n J_n(x)) = x^n J_{n-1}(x)$, $\frac{d}{dx}(x^{-n} J_n(x)) = -x^{-n} J_{n+1}(x)$).

Section-III

Marks=30

1. From

$$J_\nu(x) = \frac{1}{2\pi i} \left(\frac{x}{2}\right)^\nu \int_{-1}^1 t^{-\nu-1} e^{itx/2} dt,$$

derive the recurrence relation

$$\frac{d}{dx} J_\nu(x) = \frac{\nu}{x} J_\nu(x) - J_{\nu+1}(x).$$

2. Find the Green's function for the operator

$$L_y(x) = \frac{d}{dx} \left(x \frac{dy(x)}{dx} \right) - \frac{n^2}{x} y(x), \quad y(0) \rightarrow \text{finite and } y(1) = 0.$$

3. Show that

$$\int_0^\infty e^{-x} L_l(x) L_m(x) dx = \delta_{lm},$$

where $L_l(x)$ and $L_m(x)$ are Laguerre's polynomials.

Solve the wave equation

$$\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y(x,t)}{\partial t^2}, \quad \text{with initial condition } y(x,0) = f(x),$$

using Fourier transformation.

Prove that

$$|\Gamma(\alpha + i\beta)| = |\Gamma(\alpha)| \prod_{n=0}^{\infty} \left[1 + \frac{\beta^2}{(\alpha + n)^2} \right]^{-1/2}$$